

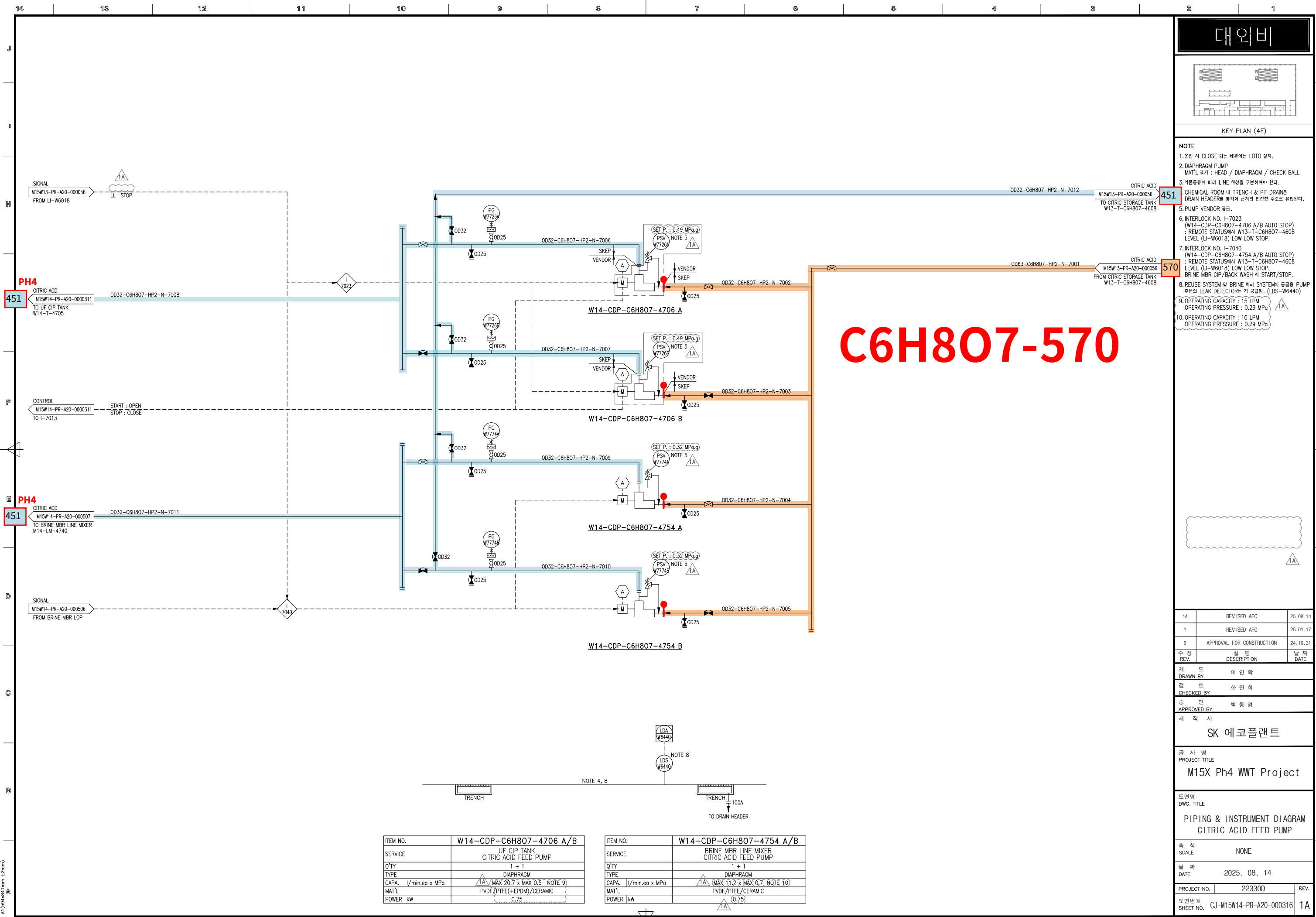
DRAWING LIST FOR TEST PACKAGE

PROJECT NO.: 22328D

PROJECT TITLE : M15X Ph2 Project

C6H8O7-570

| No | PKG No | UNIT | LINE NUMBER               |        |     | P&ID                    |     | REMARK |
|----|--------|------|---------------------------|--------|-----|-------------------------|-----|--------|
|    |        |      | ISO DRAWING No            | SHEET  | REV | P&ID No                 | REV |        |
| 1  | 570    | 4FA  | OD32-C6H8O7-HP2-N-7002-01 | 1 OF 1 | 0   | CJ-M15W14-PR-A20-000316 | 1A  |        |
| 2  | 570    | 4FA  | OD32-C6H8O7-HP2-N-7003-01 | 1 OF 1 | 0   | CJ-M15W14-PR-A20-000316 | 1A  |        |
| 3  | 570    | 4FA  | OD32-C6H8O7-HP2-N-7004-01 | 1 OF 1 | 0   | CJ-M15W14-PR-A20-000316 | 1A  |        |
| 4  | 570    | 4FA  | OD32-C6H8O7-HP2-N-7005-01 | 1 OF 1 | 0   | CJ-M15W14-PR-A20-000316 | 1A  |        |
| 5  | 570    | 4FA  | OD63-C6H8O7-HP2-N-7001-01 | 1 OF 1 | 0   | CJ-M15W13-PR-A20-000056 | 2   |        |
|    |        |      |                           |        |     |                         |     |        |
|    |        |      |                           |        |     |                         |     |        |
|    |        |      |                           |        |     |                         |     |        |
|    |        |      |                           |        |     |                         |     |        |
|    |        |      |                           |        |     |                         |     |        |
|    |        |      |                           |        |     |                         |     |        |
|    |        |      |                           |        |     |                         |     |        |
|    |        |      |                           |        |     |                         |     |        |
|    |        |      |                           |        |     |                         |     |        |
|    |        |      |                           |        |     |                         |     |        |
|    |        |      |                           |        |     |                         |     |        |
|    |        |      |                           |        |     |                         |     |        |
|    |        |      |                           |        |     |                         |     |        |
|    |        |      |                           |        |     |                         |     |        |
|    |        |      |                           |        |     |                         |     |        |
|    |        |      |                           |        |     |                         |     |        |



대외비

KEY PLAN (4F)

**NOTE**

1. 운전 시 CLOSE 되는 배관에는 LOTO 설치.

2. DIAPHRAGM PUMP  
MAT'L 표기 : HEAD / DIAPHRAGM / CHECK BALL

3. 악플종류에 따라 LINE 색상을 구분하여야 한다.

4. CHEMICAL ROOM 내 TRENCH & PIT DRAIN  
DRAIN HEADER를 통하여 근처의 인접한 수조로 유입된다.

5. PUMP VENDOR 공급.

6. INTERLOCK NO. I-7023  
(W14-CDP-C6H807-4706 A/B AUTO STOP)  
: REMOTE STATUS에서 W13-T-C6H807-4608  
LEVEL (LI-W6018) LOW LOW STOP.

7. INTERLOCK NO. I-7040  
(W14-CDP-C6H807-4754 A/B AUTO STOP)  
: REMOTE STATUS에서 W13-T-C6H807-4608  
LEVEL (LI-W6018) LOW LOW STOP.  
BRINE MBR CIP/BACK WASH 시 START/STOP.

8. REUSE SYSTEM 및 BRINE 처리 SYSTEM의 공급용 PUMP  
주변의 LEAK DETECTOR는 기 공급됨. (LDS-W6440)

9. OPERATING CAPACITY : 15 LPM  
OPERATING PRESSURE : 0.29 MPa

10. OPERATING CAPACITY : 10 LPM  
OPERATING PRESSURE : 0.29 MPa

|                                                                           |                           |            |
|---------------------------------------------------------------------------|---------------------------|------------|
| 1A                                                                        | REVISED AFC               | 25.08.14   |
| 1                                                                         | REVISED AFC               | 25.01.17   |
| 0                                                                         | APPROVAL FOR CONSTRUCTION | 24.10.31   |
| 수정<br>REV.                                                                | 설명<br>DESCRIPTION         | 날짜<br>DATE |
| 제도이인학<br>DRAWN BY                                                         |                           |            |
| 검토헌진희<br>CHECKED BY                                                       |                           |            |
| 승인박동명<br>APPROVED BY                                                      |                           |            |
| 제작사<br>SK 에코플랜트                                                           |                           |            |
| 공사명<br>PROJECT TITLE<br>M15X Ph4 WWT Project                              |                           |            |
| 도면명<br>DWG. TITLE<br>PIPING & INSTRUMENT DIAGRAM<br>CITRIC ACID FEED PUMP |                           |            |
| 축척<br>SCALE<br>NONE                                                       |                           |            |
| 날짜<br>DATE<br>2025. 08. 14                                                |                           |            |
| PROJECT NO.                                                               | 22330D                    | REV.       |
| 도면번호<br>SHEET NO.                                                         | CJ-M15W14-PR-A20-000316   | 1A         |

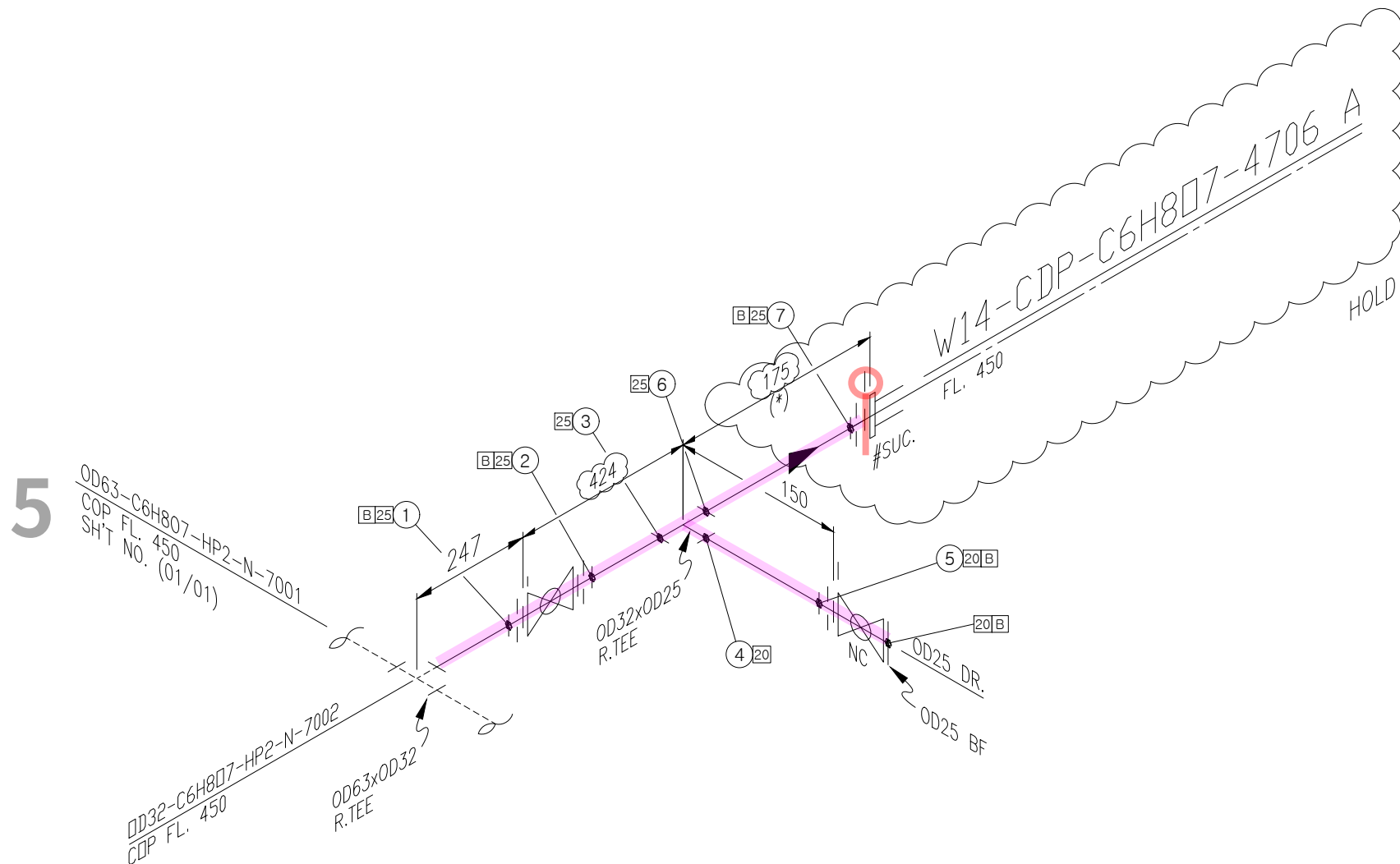
|           |                                                          |
|-----------|----------------------------------------------------------|
| ITEM NO.  | W14-CDP-C6H807-4706 A/B                                  |
| SERVICE   | UF CIP TANK<br>CITRIC ACID FEED PUMP                     |
| Q'TY      | 1 + 1                                                    |
| TYPE      | DIAPHRAGM                                                |
| CAPA.     | 1/min.ea x MPa                                           |
| MAT'L     | (MAX 20.7 x MAX 0.5" NOTE 9)<br>PVDF/PTFE(+EPDM)/CERAMIC |
| POWER  kW | 0.75                                                     |

|           |                                                    |
|-----------|----------------------------------------------------|
| ITEM NO.  | W14-CDP-C6H807-4754 A/B                            |
| SERVICE   | BRINE MBR LINE MIXER<br>CITRIC ACID FEED PUMP      |
| Q'TY      | 1 + 1                                              |
| TYPE      | DIAPHRAGM                                          |
| CAPA.     | 1/min.ea x MPa                                     |
| MAT'L     | (MAX 11.2 x MAX 0.7" NOTE 10)<br>PVDF/PTFE/CERAMIC |
| POWER  kW | 0.75                                               |





| ITEM | SIZE      | Q'TY | DESCRIPTION                                        |
|------|-----------|------|----------------------------------------------------|
| HP2  |           |      |                                                    |
| P    | OD32      | 0.9M | PIPE, HDPE PE100, PE, SOCKET FUSION, PN16/SDR11    |
| P    | OD25      | 0.2M | PIPE, HDPE PE100, PE, SOCKET FUSION, PN16/SDR11    |
| TR   | OD32xOD25 | 1EA  | TEE RED, HDPE PE100, PE, SOCKET FUSION, PN16/SDR11 |
| F    | OD32      | 3EA  | FLG LAP JOINT / STUB END SET, KS 10K, FF,          |
|      |           |      | LAP JOINT-ST3304 / STUB END-HDPE PE100             |
| F    | OD25      | 1EA  | FLG LAP JOINT / STUB END SET, KS 10K, FF,          |
|      |           |      | LAP JOINT-ST3304 / STUB END-HDPE PE100             |
| FB   | OD25      | 1EA  | FLG BLIND & LINER SET, KS 10K, FF, BLIND-ST3304    |
|      |           |      | / LINDER-HDPE PE100 (MIN 5.0MM TH'K)               |
| G    | OD32      | 3EA  | GASKET, KS 10K, FF, PTFE+EPDM, 3.0MM TH'K          |
| G    | OD25      | 1EA  | GASKET, KS 10K, FF, PTFE+EPDM, 3.0MM TH'K          |
| G2   | OD25      | 1EA  | GASKET, KS 10K, FF, PTFE+EPDM, BLIND, 3.0MM TH'K   |
| B    | OD32      | 3SET | MACHINE BOLT & NUTS :STS 304 / W WASHERS           |
| B    | OD25      | 2SET | MACHINE BOLT & NUTS :STS 304 / W WASHERS           |
| VB   | OD32      | 1EA  | BALL VALVE, KS 10K, FF FLG'D, BODY-PP /            |
|      |           |      | TRIM-PP / SEAT-PTFE, FB, LEVER                     |
| VB   | OD25      | 1EA  | BALL VALVE, KS 10K, FF FLG'D, BODY-PP /            |
|      |           |      | TRIM-PP / SEAT-PTFE, FB, LEVER                     |
| LOTO | OD25      | 1EA  | LOTO                                               |

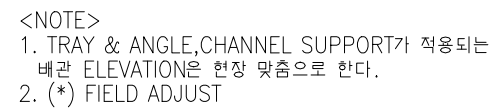


<NOTE>

1. TRAY & ANGLE, CHANNEL SUPPORT가 적용되는 배관 ELEVATION은 현장 맞춤으로 한다.
2. (\*) FIELD ADJUST

|                           |                           |                                |                        |                                |                                 |                                           |                         |                                    |                          |                                              |                       |                               |              |                      |                          |
|---------------------------|---------------------------|--------------------------------|------------------------|--------------------------------|---------------------------------|-------------------------------------------|-------------------------|------------------------------------|--------------------------|----------------------------------------------|-----------------------|-------------------------------|--------------|----------------------|--------------------------|
| * PIPE : P/P1             | * 90° ELBOW : E90         | * 45° ELBOW : E45              | * TEE : T              | * RED.TEE : TR                 | * STUB-IN : STU                 | * REINFORCING PAD : RP                    | * CON.RED : RC          | * ECC.RED : RE                     | * CON.SWAGE : RCS/RCS1   | * ECC.SWAGE : RES/RES1                       | * NIPPLE : NP/NP1/NP2 | * CAP : CAP/CAP1              | * PLUG : PLG | * FULL COUPLING : FC | * HALF COUPLING : HF/HF1 |
| * STUB-IN : STU           | * SOCKET(HDPE/PVC) : SC   | * SADDLE(HDPE) : SD            | * FLANGE : F/F1        | * GASKET : G/G1/G2             | * BOLT & NUT : B/B1             | * BLIND FLANGE : FB/FB1                   | * GATE VA. : VG/VG1/VG2 | * BALL VA. : VB/VB1/VB10/VB2/VB210 | * BUTTERFLY VA. : VF/VF4 | * CHECK VA. : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7 |                       |                               |              |                      |                          |
| * GLOBE VA. : VL/VL1/VL2  | * DIAPHRAGM VA. : VD/VD10 | * DAMPER : VP                  | * QUICK COUPLING : SQC | * FLEXIBLE JOINT(STEEL) : FLXB | * FLEXIBLE JOINT(RUBBER) : FLXR | * FLEXIBLE JOINT(STEEL PTFE LINED) : FLXL | * FLEXIBLE HOSE : FLXH  | * STEAM TRAP : ST                  | * SPRAY NOZZLE : SPN     | * EMERGENCY SHOWER : SSH                     |                       |                               |              |                      |                          |
| * STRAINER(Y-TYPE) : YSTR | * STRAINER(T-TYPE) : TSTR | * STRAINER(BUCKET-TYPE) : BSTR | * ANCHOR PLATE : STOP  | * BIRD SCREEN : BS             | * ISOLATION KIT : IK            | * UNLOADING BOX : UB                      | * AIR TRAP : AT         | * LOTO : LOTO                      | * SIGHT GLASS : SG       | * PRESSURE GAUGE : PG                        | * SHILED : SH         | * SPECIAL BOLT & NUT : BL/BL1 |              |                      |                          |

|            |        |   |           |                                  |               |                                             |                                                                                       |          |                           |     |     |       |                                                                                                     |                     |           |
|------------|--------|---|-----------|----------------------------------|---------------|---------------------------------------------|---------------------------------------------------------------------------------------|----------|---------------------------|-----|-----|-------|-----------------------------------------------------------------------------------------------------|---------------------|-----------|
| INSULATION | CODE   | N | CONDITION | PRESS.<br>(kg/cm <sup>2</sup> g) | TEMP.<br>(*C) | AREA W14                                    |                                                                                       |          |                           |     |     |       | CONTRACTOR<br> | PROJECT NAME        |           |
|            | THICK. | - | DESIGN    | 1.0                              | 60            | UNIT 4FA                                    |                                                                                       |          |                           |     |     |       |                                                                                                     | M15X Ph4 Project    |           |
|            |        |   | OPERATING | ATM                              | AMB           | REF. PLAN DWG.<br>CJ-M15W14-PI-A30-040009-A |                                                                                       |          |                           |     |     |       | COMMODITY C6H8O7                                                                                    | JOB NO.             |           |
|            |        |   |           |                                  |               | REF. P & ID<br>CJ-M15W14-PR-A20-000316      |  | 25.01.24 | APPROVED FOR CONSTRUCTION | 황래중 | 안남후 | 박동영   | LINE NO. OD32-C6H807-HP2-N-7002                                                                     | SH'T NO.<br>01 / 01 | REV.<br>0 |
|            |        |   |           |                                  |               |                                             | REV. NO.                                                                              | DATE     | DESCRIPTION               | BY  | CHK | APP'D |                                                                                                     |                     |           |

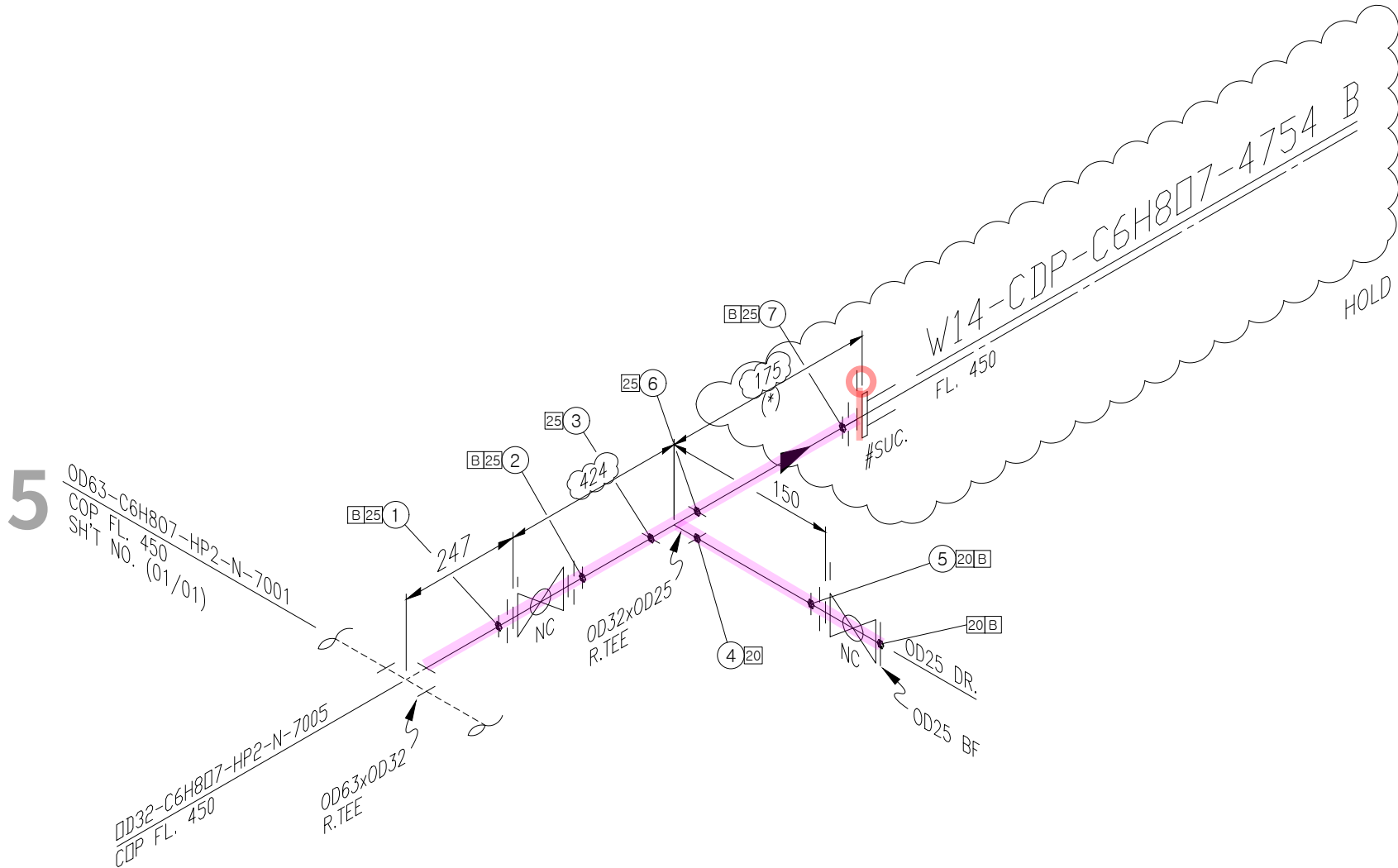
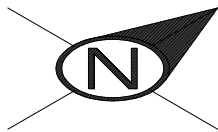


|                           |                           |                                |                        |                                |                                 |                                           |                         |                                          |                          |                                              |                       |                               |              |                      |                          |
|---------------------------|---------------------------|--------------------------------|------------------------|--------------------------------|---------------------------------|-------------------------------------------|-------------------------|------------------------------------------|--------------------------|----------------------------------------------|-----------------------|-------------------------------|--------------|----------------------|--------------------------|
| * PIPE : P/P1             | * 90° ELBOW : E90         | * 45° ELBOW : E45              | * TEE : T              | * RED.TEE : TR                 | * STUB-IN : STU                 | * REINFORCING PAD : RP                    | * CON.RED : RC          | * ECC.RED : RE                           | * CON.SWAGE : RCS/RCS1   | * ECC.SWAGE : RES/RES1                       | * NIPPLE : NP/NP1/NP2 | * CAP : CAP/CAP1              | * PLUG : PLG | * FULL COUPLING : FC | * HALF COUPLING : HF/HF1 |
| * STUB-IN : STU           | * SOCKET(HDPE/PVC) : SC   | * SADDLE(HDPE) : SD            | * FLANGE : F/F1        | * GASKET : G/G1/G2             | * BOLT & NUT : B/B1             | * BLIND FLANGE : FB/FB1                   | * GATE VA. : VG/VG1/VG2 | * BALL VA. : VB/VBLO/VB1/VB1LO/VB2/VB2LO | * BUTTERFLY VA. : VF/VF4 | * CHECK VA. : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7 |                       |                               |              |                      |                          |
| * GLOBE VA. : VL/VL1/VL2  | * DIAPHRAGM VA. : VD/VDLO | * DAMPER : VP                  | * QUICK COUPLING : SQC | * FLEXIBLE JOINT(STEEL) : FLXB | * FLEXIBLE JOINT(RUBBER) : FLXR | * FLEXIBLE JOINT(STEEL PTFE LINED) : FLXL | * FLEXIBLE HOSE : FLXH  | * STEAM TRAP : ST                        | * SPRAY NOZZLE : SPN     | * EMERGENCY SHOWER : SSH                     |                       |                               |              |                      |                          |
| * STRAINER(Y-TYPE) : YSTR | * STRAINER(T-TYPE) : TSTR | * STRAINER(BUCKET-TYPE) : BSTR | * ANCHOR PLATE : STOP  | * BIRD SCREEN : BS             | * ISOLATION KIT : IK            | * UNLOADING BOX : UB                      | * AIR TRAP : AT         | * LOTO : LOTO                            | * SIGHT GLASS : SG       | * PRESSURE GAUGE : PG                        | * SHILED : SH         | * SPECIAL BOLT & NUT : BL/BL1 |              |                      |                          |

|            |        |   |           |                                  |               |                           |     |                                                                                       |          |                           |     |     |       |                                                                                       |                  |      |
|------------|--------|---|-----------|----------------------------------|---------------|---------------------------|-----|---------------------------------------------------------------------------------------|----------|---------------------------|-----|-----|-------|---------------------------------------------------------------------------------------|------------------|------|
| INSULATION | CODE   | N | CONDITION | PRESS.<br>(kg/cm <sup>2</sup> g) | TEMP.<br>(°C) | AREA                      | W14 |                                                                                       |          |                           |     |     |       | CONTRACTOR                                                                            | PROJECT NAME     |      |
|            | THICK. | — | DESIGN    | 1.0                              | 60            | UNIT                      | 4FA |                                                                                       |          |                           |     |     |       |  | M15X Ph4 Project |      |
|            |        |   | OPERATING | ATM                              | AMB           | REF. PLAN DWG.            |     |                                                                                       |          |                           |     |     |       | COMMODITY                                                                             | JOB NO.          |      |
|            |        |   |           |                                  |               | CJ-M15W14-PI-A30-040009-A |     |                                                                                       |          |                           |     |     |       | C6H807                                                                                | 22330D           |      |
|            |        |   |           |                                  |               | REF. P & ID               |     |  | 25.01.24 | APPROVED FOR CONSTRUCTION | 황래중 | 안남후 | 박동영   | LINE NO.                                                                              | SH'T NO.         | REV. |
|            |        |   |           |                                  |               | CJ-M15W14-PR-A20-000316   |     | REV. NO.                                                                              | DATE     | DESCRIPTION               | BY  | CHK | APP'D | 0D32-C6H807-HP2-N-7003                                                                | 01 / 01          | 0    |







<NOTE>  
1. TRAY & ANGLE,CHANNEL SUPPORT가 적용되는  
배관 ELEVATION은 현장 맞춤으로 한다.  
2. (\*) FIELD ADJUST

\* PIPE : P/P1 \* 90° ELBOW : E90 \* 45° ELBOW : E45 \* TEE : T \* RED.TEE : TR \* STUB-IN : STU \* REINFORCING PAD : RP \* CON.RED : RC \* ECC.RED : RE \* CON.SWAGE : RCS/RCS1 \* ECC.SWAGE : RES/RES1 \* NIPPLE : NP/NP1/NP2 \* CAP : CAP/CAP1 \* PLUG : PLG \* FULL COUPLING : FC \* HALF COUPLING : HF/HF1  
\* STUB-IN : STU \* SOCKET(HDPE/PVC) : SC \* SADDLE(HDPE) : SD \* FLANGE : F/F1 \* GASKET : G/G1/G2 \* BOLT & NUT : B/B1 \* BLIND FLANGE : FB/FB1 \* GATE VA. : VG/VG1/VG2 \* BALL VA. : VB/VBLO/VB1/VB1LO/VB2/VB2LO \* BUTTERFLY VA. : VF/VF4 \* CHECK VA. : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7  
\* GLOBE VA. : VL/VL1/VL2 \* DIAPHRAGM VA. : VD/VDLO \* DAMPER : VP \* QUICK COUPLING : SQC \* FLEXIBLE JOINT(STEEL) : FLXB \* FLEXIBLE JOINT(RUBBER) : FLXR \* FLEXIBLE JOINT(STEEL PTFE LINED) : FLXL \* FLEXIBLE HOSE : FLXH \* STEAM TRAP : ST \* SPRAY NOZZLE : SPN \* EMERGENCY SHOWER : SSH  
\* STRAINER(Y-TYPE) : YSTR \* STRAINER(T-TYPE) : TSTR \* STRAINER(BUCKET-TYPE) : BSTR \* ANCHOR PLATE : STOP \* BIRD SCREEN : BS \* ISOLATION KIT : IK \* UNLOADING BOX : UB \* AIR TRAP : AT \* LOTO : LOTO \* SIGHT GLASS : SG \* PRESSURE GAUGE : PG \* SHILED : SH \* SPECIAL BOLT & NUT : BL/BL1



| ITEM | SIZE      | Q'TY | DESCRIPTION                                        |
|------|-----------|------|----------------------------------------------------|
| HP2  |           |      |                                                    |
| P    | OD32      | 0.9M | PIPE, HDPE PE100, PE, SOCKET FUSION, PN16/SDR11    |
| P    | OD25      | 0.2M | PIPE, HDPE PE100, PE, SOCKET FUSION, PN16/SDR11    |
| TR   | OD32xOD25 | 1EA  | TEE RED, HDPE PE100, PE, SOCKET FUSION, PN16/SDR11 |
| F    | OD32      | 3EA  | FLG LAP JOINT / STUB END SET, KS 10K, FF,          |
|      |           |      | LAP JOINT-ST304 / STUB END-HDPE PE100              |
| F    | OD25      | 1EA  | FLG LAP JOINT / STUB END SET, KS 10K, FF,          |
|      |           |      | LAP JOINT-ST304 / STUB END-HDPE PE100              |
| FB   | OD25      | 1EA  | FLG BLIND & LINER SET, KS 10K, FF, BLIND-ST304     |
|      |           |      | / LINDER-HDPE PE100 (MIN 5.0MM TH'K)               |
| G    | OD32      | 3EA  | GASKET, KS 10K, FF, PTFE+EPDM, 3.0MM TH'K          |
| G    | OD25      | 1EA  | GASKET, KS 10K, FF, PTFE+EPDM, 3.0MM TH'K          |
| G2   | OD25      | 1EA  | GASKET, KS 10K, FF, PTFE+EPDM, BLIND, 3.0MM TH'K   |
| B    | OD32      | 3SET | MACHINE BOLT & NUTS :STS 304 / W WASHERS           |
| B    | OD25      | 2SET | MACHINE BOLT & NUTS :STS 304 / W WASHERS           |
| VB   | OD32      | 1EA  | BALL VALVE, KS 10K, FF FLG'D, BODY-PP /            |
|      |           |      | TRIM-PP / SEAT-PTFE, FB, LEVER                     |
| VB   | OD25      | 1EA  | BALL VALVE, KS 10K, FF FLG'D, BODY-PP /            |
|      |           |      | TRIM-PP / SEAT-PTFE, FB, LEVER                     |
| LOTO | OD25      | 1EA  | LOTO                                               |
| LOTO | OD32      | 1EA  | LOTO                                               |

CLIENT

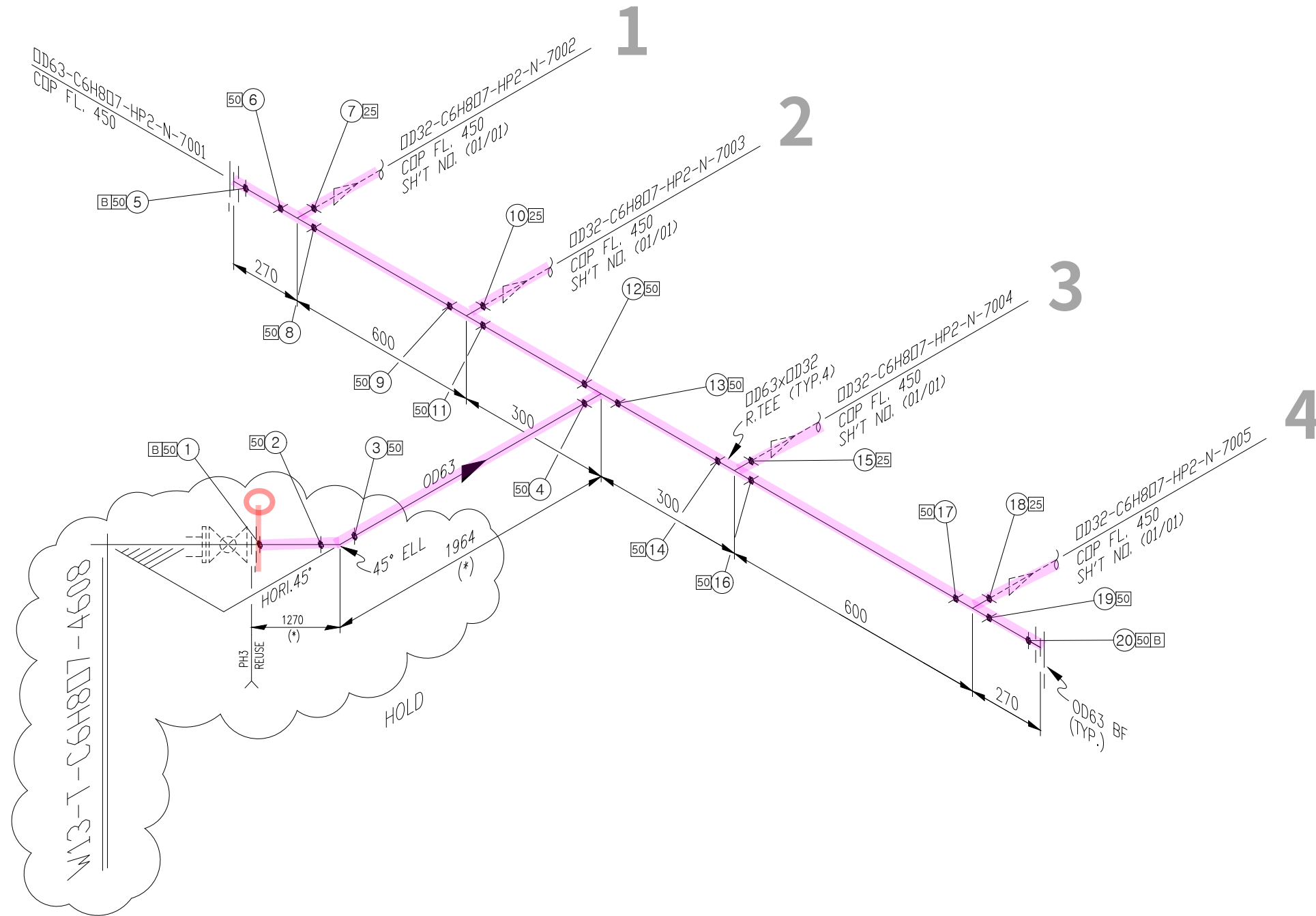


4

|                        |  |                  |      |
|------------------------|--|------------------|------|
| CONTRACTOR             |  | PROJECT NAME     |      |
| SK ecoplant            |  | M15X Ph4 Project |      |
| COMMODITY              |  | JOB NO.          |      |
| C6H807                 |  | 22330D           |      |
| LINE NO.               |  | SH'T NO.         | REV. |
| OD32-C6H807-HP2-N-7005 |  | 01 / 01          | 0    |



| ITEM | SIZE      | Q'TY | DESCRIPTION                                          |
|------|-----------|------|------------------------------------------------------|
| HP2  |           |      |                                                      |
| P    | OD63      | 5.2M | PIPE, HDPE PE100, PE, SOCKET FUSION, PN16/SDR11      |
| E45  | OD63      | 1EA  | ELBOW 45 DEG, HDPE PE100, PE, SOCKET FUSION,         |
| T    | OD63      | 1EA  | TEE EQUAL, HDPE PE100, PE, SOCKET FUSION, PN16/SDR11 |
| TR   | OD63xOD32 | 4EA  | TEE RED, HDPE PE100, PE, SOCKET FUSION, PN16/SDR11   |
| F    | OD63      | 3EA  | FLG LAP JOINT / STUB END SET, KS 10K, FF,            |
|      |           |      | LAP JOINT-STS304 / STUB END-HDPE PE100               |
| FB   | OD63      | 2EA  | FLG BLIND & LINER SET, KS 10K, FF, BLIND-STS304      |
|      |           |      | / LINDER-HDPE PE100 (MIN 5.0MM TH'K)                 |
| G    | OD63      | 1EA  | GASKET, KS 10K, FF, PTFE+EPDM, 3.0MM TH'K            |
| G2   | OD63      | 2EA  | GASKET, KS 10K, FF, PTFE+EPDM, BLIND, 3.0MM TH'K     |
| B    | OD63      | 3SET | MACHINE BOLT & NUTS :STS 304 / W WASHERS             |



<NOTE>  
1. TRAY & ANGLE, CHANNEL SUPPORT가 적용되는  
배관 ELEVATION은 현장 맞춤으로 한다.  
2. (\*) FIELD ADJUST

|                           |                           |                                |                        |                                |                                 |                                           |                         |                                    |                          |                                              |                       |                               |              |                      |                          |
|---------------------------|---------------------------|--------------------------------|------------------------|--------------------------------|---------------------------------|-------------------------------------------|-------------------------|------------------------------------|--------------------------|----------------------------------------------|-----------------------|-------------------------------|--------------|----------------------|--------------------------|
| * PIPE : P/P1             | * 90° ELBOW : E90         | * 45° ELBOW : E45              | * TEE : T              | * RED.TEE : TR                 | * STUB-IN : STU                 | * REINFORCING PAD : RP                    | * CON.RED : RC          | * ECC.RED : RE                     | * CON.SWAGE : RCS/RCS1   | * ECC.SWAGE : RES/RES1                       | * NIPPLE : NP/NP1/NP2 | * CAP : CAP/CAP1              | * PLUG : PLG | * FULL COUPLING : FC | * HALF COUPLING : HF/HF1 |
| * STUB-IN : STU           | * SOCKET(HDPE/PVC) : SC   | * SADDLE(HDPE) : SD            | * FLANGE : F/F1        | * GASKET : G/G1/G2             | * BOLT & NUT : B/B1             | * BLIND FLANGE : FB/FB1                   | * GATE VA. : VG/VG1/VG2 | * BALL VA. : VB/VB1/VB10/VB2/VB210 | * BUTTERFLY VA. : VF/VF4 | * CHECK VA. : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7 |                       |                               |              |                      |                          |
| * GLOBE VA. : VL/VL1/VL2  | * DIAPHRAGM VA. : VD/VD10 | * DAMPER : VP                  | * QUICK COUPLING : SQC | * FLEXIBLE JOINT(STEEL) : FLXB | * FLEXIBLE JOINT(RUBBER) : FLXR | * FLEXIBLE JOINT(STEEL PTFE LINED) : FLXL | * FLEXIBLE HOSE : FLXH  | * STEAM TRAP : ST                  | * SPRAY NOZZLE : SPN     | * EMERGENCY SHOWER : SSH                     |                       |                               |              |                      |                          |
| * STRAINER(Y-TYPE) : YSTR | * STRAINER(T-TYPE) : TSTR | * STRAINER(BUCKET-TYPE) : BSTR | * ANCHOR PLATE : STOP  | * BIRD SCREEN : BS             | * ISOLATION KIT : IK            | * UNLOADING BOX : UB                      | * AIR TRAP : AT         | * LOTO : LOTO                      | * SIGHT GLASS : SG       | * PRESSURE GAUGE : PG                        | * SHILED : SH         | * SPECIAL BOLT & NUT : BL/BL1 |              |                      |                          |

|                                                                                       |  |  |  |                                                                                                                                                                             |      |
|---------------------------------------------------------------------------------------|--|--|--|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|
| CLIENT                                                                                |  |  |  |   |      |
| CONTRACTOR                                                                            |  |  |  | PROJECT NAME                                                                                                                                                                |      |
|  |  |  |  | M15X Ph4 Project                                                                                                                                                            |      |
| COMMODITY                                                                             |  |  |  | JOB NO.                                                                                                                                                                     |      |
| C6H807                                                                                |  |  |  | 22330D                                                                                                                                                                      |      |
| LINE NO.                                                                              |  |  |  | SH'T NO.                                                                                                                                                                    | REV. |
| 0D63-C6H807-HP2-N-7001                                                                |  |  |  | 01 / 01                                                                                                                                                                     | 0    |

|            |        |   |           |                     |               |                           |     |                                                                                       |          |                           |     |     |       |                                                                                       |                        |  |              |      |
|------------|--------|---|-----------|---------------------|---------------|---------------------------|-----|---------------------------------------------------------------------------------------|----------|---------------------------|-----|-----|-------|---------------------------------------------------------------------------------------|------------------------|--|--------------|------|
| INSULATION | CODE   | N | CONDITION | PRESS.<br>(kg/cm2g) | TEMP.<br>(°C) | AREA                      | W14 |                                                                                       |          |                           |     |     |       |  | CONTRACTOR             |  | PROJECT NAME |      |
|            | THICK. | — | DESIGN    | 1.0                 | 60            | UNIT                      | 4FA |                                                                                       |          |                           |     |     |       |                                                                                       | M15X Ph4 Project       |  |              |      |
|            |        |   | OPERATING | ATM                 | AMB           | REF. PLAN DWG.            |     |                                                                                       |          |                           |     |     |       | COMMODITY                                                                             | C6H807                 |  | JOB NO.      |      |
|            |        |   |           |                     |               | CJ-M15W14-PI-A30-040009-A |     |  | 25.01.24 | APPROVED FOR CONSTRUCTION | 황래중 | 안남후 | 박동영   |                                                                                       | 22330D                 |  |              |      |
|            |        |   |           |                     |               | REF. P & ID               |     | REV. NO.                                                                              | DATE     | DESCRIPTION               | BY  | CHK | APP'D | LINE NO.                                                                              | 0D63-C6H807-HP2-N-7001 |  | SH'T NO.     | REV. |
|            |        |   |           |                     |               | CJ-M15W14-PR-A20-000316   |     |                                                                                       |          |                           |     |     |       |                                                                                       |                        |  | 01 / 01      | 0    |